Module 2: Basic Chemistry - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. Table salt (NaCl) forms when a sodium atom gives up an electron to a chlorine atom. What kind of bond holds the result together?
    - A. A covalent bond from shared electrons
    - B. An ionic bond between oppositely charged ions
    - C. A hydrogen bond
    - D. No bond at all
    - Answer: B
    - Why: Transferring electrons creates charged ions that attract each other, which is an ionic bond. Sharing electrons instead would make a covalent bond.
2. Why can water dissolve so many substances important to cells?
    - A. Water is polar, so its partial charges pull on charged and polar molecules.
    - B. Water carries no charge anywhere in the molecule.
    - C. Water is a pure element.
    - D. Water is always strongly acidic.
    - Answer: A
    - Why: Water's polarity (uneven charge distribution) lets it surround and separate ions and polar molecules, so they dissolve.
3. A solution changes from pH 7 to pH 4. This means it became:
    - A. more basic, with fewer hydrogen ions
    - B. exactly neutral
    - C. a pure element
    - D. more acidic, with more hydrogen ions
    - Answer: D
    - Why: Lower pH means a higher concentration of hydrogen ions, which is more acidic.

4. Blood keeps a nearly steady pH even when acids enter it. What best explains this stability?
    - A. Blood contains no water.
    - B. Acids are unable to enter blood.
    - C. A buffer resists pH change by absorbing or releasing hydrogen ions.
    - D. pH never changes in any liquid.
    - Answer: C
    - Why: Buffers stabilize pH by taking up or releasing hydrogen ions, which protects enzymes and cell function.